WRF Model and Meteosat with Machine Learning for Forecast of Low Stratus and Fog Tested on a Special Event

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1 Introduction and objectives

Fog and very low stratus clouds are among the most disruptive weather phenomena, yet they remain difficult to forecast (Haeffelin et al., 2010). A rare multi-day fog/low-stratus event in the eastern Mediterranean region (3–6 January 2021) caused widespread visibility reductions and even forced a temporary closure of airports.

We investigate whether combining high-resolution numerical weather prediction (NWP) with satellite observations and machine learning (ML) can improve the prediction of such events. Our study uses the Weather Research and Forecasting model with real-time four-dimensional data assimilation (WRF-RTFDDA; Huang et al., 2018), together with Meteosat SEVIRI satellite imagery and an ML-based post-processing approach, to forecast the timing and spatial extent of the January 2021 fog/low stratus outbreak. This approach exemplifies a climate informatics paradigm, integrating physical modeling, remote sensing data, and ML techniques to enhance predictions of fog and low cloud occurrence.

2 Methods and data

For this study, we used the WRF-RTFDDA model (Huang et al., 2018) with 1-km horizontal resolution over Israel to generate high-resolution weather forecasts. Meteosat Second Generation (MSG) SEVIRI satellite imagery (the EUMETSAT "night fog" RGB product at night and the daytime natural-color "fog" composite) was processed to produce binary masks of fog/very-low-cloud presence, serving as both qualitative truth and ML training data. We also gathered surface visibility observations from five weather stations in central and southern Israel, which provide independent ground truth for fog (defined as visibility ≤ 1 km).

Two post-processing methods were applied to the WRF output to identify low stratus/fog areas. The first is a Threshold Algorithm (TA) that flags fog/low stratus at a model grid point if near-surface conditions indicate saturation and calm winds (e.g., dewpoint depression ≤ ~2 °C, high relative humidity, and light winds). The second method is an ML classifier using gradient-boosted trees (GBT). The GBT model was trained to replicate the satellite-detected fog/low cloud mask from the corresponding WRF predictors (near-surface temperature, humidity, and wind). Given the limited dataset (four nights), we adopted a leave-one-day-out (four-fold) cross-validation approach: the GBT was trained on three nights and tested on the remaining night in each fold (Kim et al., 2022). This ensured that the ML predictions for each night were based only on information from the other days. Finally, the fog/low cloud forecasts from both methods were compared to the station observations to calculate performance metrics (precision, recall, F1-score).

3 Preliminary results and/or expected outcomes

Both the TA and ML post-processing approaches successfully reproduced the broad spatial and temporal evolution of the low stratus/fog event. Satellite imagery showed low clouds forming over the southern coastal plain each night (Figure 2) and spreading northward by morning; the WRF-based forecasts captured this progression and even reflected smaller-scale features like clear patches over urban areas (likely due to heat-island effects). When evaluated against the surface visibility stations, the fog/low cloud forecasts achieved an F1-score of approximately 0.65, indicating moderate predictive skill for this event.

The two methods performed similarly: the TA approach was slightly more conservative (fewer false alarms, higher precision) while the ML classifier was slightly more aggressive (detecting more fog occurrences, higher recall). In practice, the simple rule-based diagnostic performed nearly as well as the data-driven model, suggesting that the chosen meteorological predictors encapsulate much of the fog signal. The ML approach, however, added value by better matching the satellite-observed cloud structures. Moreover, the WRF model's three-dimensional output provides insight into whether a low cloud is at the surface (fog) or just above it (low stratus)—a distinction satellite data alone cannot make. Overall, this integrated modeling–satellite–ML system proved capable of providing useful guidance on fog and low cloud occurrence. Future applications of this framework to additional fog events, potentially with expanded predictors, are expected to further enhance fog forecasting in the region.

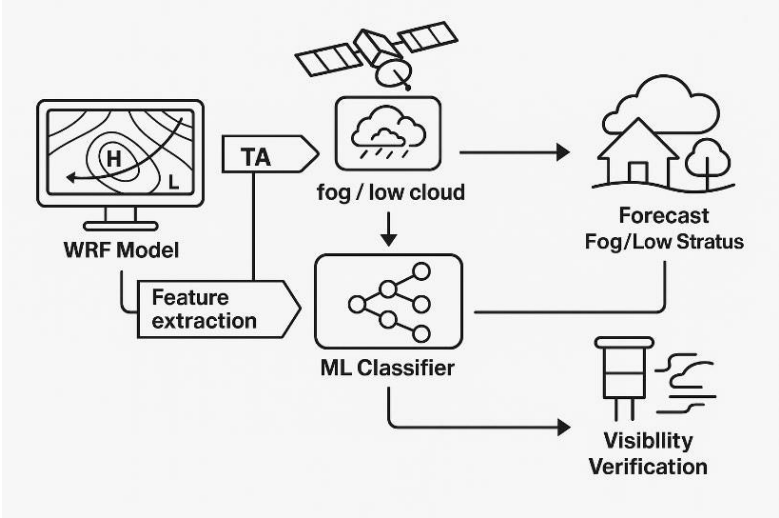

Figure 1. Schematic of the modeling, satellite processing, and machine learning workflow for forecasting low stratus and fog. The workflow integrates high-resolution WRF model forecasts with Meteosat satellite-derived fog/low cloud detection and ML post-processing. The ML classifier is trained on satellite-based labels and applied to WRF output features, while the simpler TA method uses physical criteria on the WRF fields to diagnose fog/low stratus.

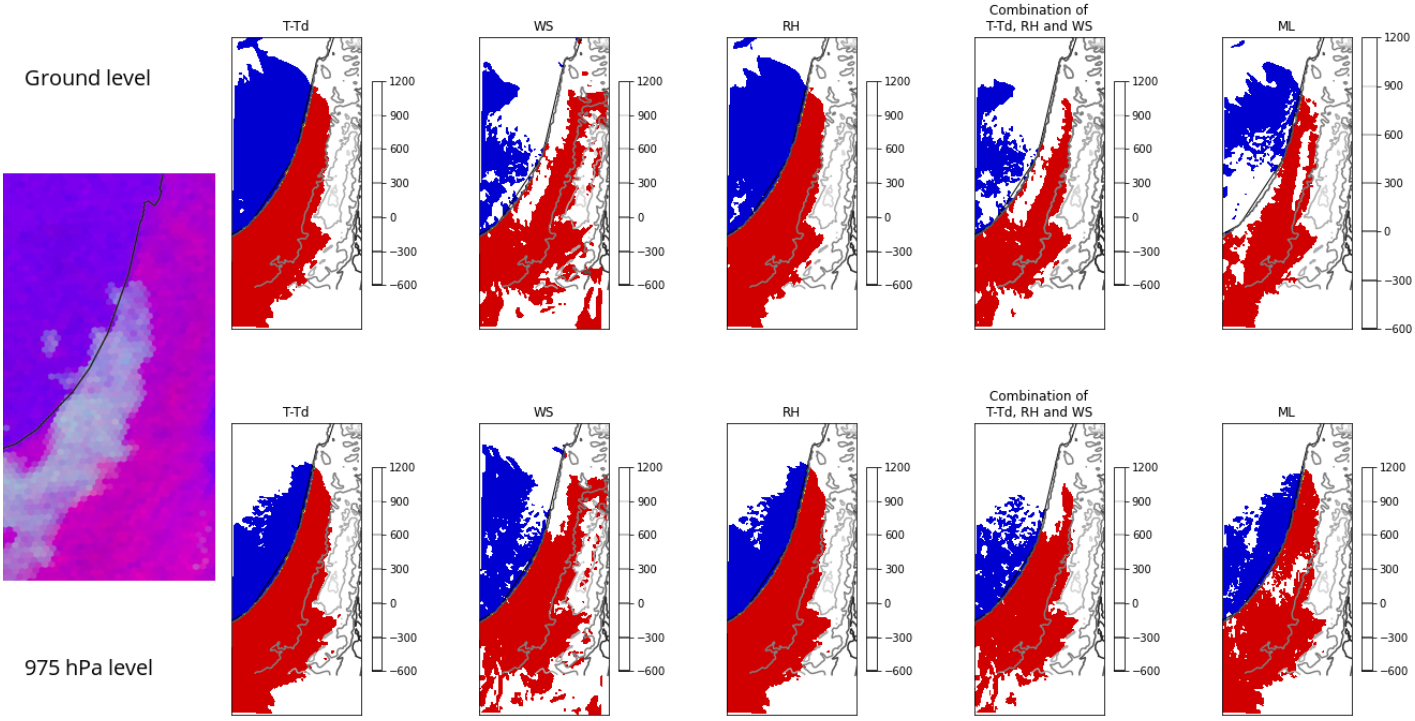

Figure 2. RGB images of fog/low clouds from EUMETSAT data (1st column from left) for nighttime; maps from WRF model of T-Td, wind and relative humidity (2nd, 3rd and 4th columns from the left). The 5th column shows the overlap of T-Td with RH and WS. The 6th column is the GBT result. All calculated maps are overlapped on terrain height. Upper row calculations: ground level. Lower row calculations: 975 hPa level. All panels are valid at 00 UTC 0f the 6th of January, 2021.

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